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Che et al.

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(54) **JIGSAW PUZZLE TABLE**
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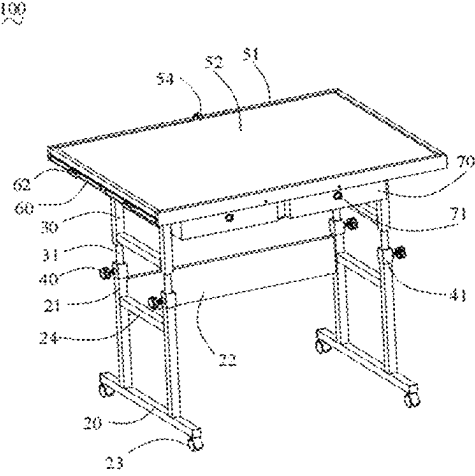
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A47B 13/08 (2006.01)
A47B 25/00 (2006.01)
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(56) **References Cited**
U.S. PATENT DOCUMENTS
129,870 A * 7/1872 Stahn A47B 27/02 108/10
702,749 A * 6/1902 Smith A47B 27/02 108/10

(Continued)
Primary Examiner — Jose V Chen

(57) **ABSTRACT**
A jigsaw puzzle table includes: a puzzle board; a board bracket, a lower support, and at least one bracket storage member. The puzzle board is mounted on a mounting end of the board bracket. The board bracket is mounted between the puzzle board and the lower support, the board bracket and the lower support are configured to support the puzzle board cooperatively. The at least one bracket storage member is configured to store puzzle pieces, and the at least one bracket storage member is supported by and movably mounted on the board bracket.

20 Claims, 13 Drawing Sheets



Related U.S. Application Data

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Des. 1,029,113.

(56)

References Cited

U.S. PATENT DOCUMENTS

922,565 A *	5/1909	Cole	A47B 27/00	4,312,507 A *	1/1982	Smith	A63F 3/0023
			211/12				273/287
1,203,659 A *	11/1916	Smith	A47B 23/043	4,865,325 A *	9/1989	Stolz	A63F 9/1044
			248/456				273/157 R
2,708,808 A *	5/1955	Simpson	A47B 19/06	5,529,173 A *	6/1996	Salacuse	B65D 75/22
			312/316				206/315.1
3,026,158 A *	3/1962	Freeman	A47B 5/04	D400,600 S *	11/1998	Ackerman	D21/478
			108/134	6,045,108 A *	4/2000	Cziraky	A45C 9/00
3,123,935 A *	3/1964	Williams	A47B 13/16				248/461
			248/500	7,073,449 B2 *	7/2006	Pipkin	B64D 11/0638
3,125,971 A *	3/1964	Wetherill	A47B 25/00				108/9
			108/125	7,552,821 B1 *	6/2009	Demers	A63F 9/1044
3,682,109 A *	8/1972	Gordon	A47B 13/081				206/499
			108/9	8,322,290 B1 *	12/2012	Mignano	A47B 23/043
3,773,154 A *	11/1973	Mitchell	A47B 23/043				248/456
			248/455	8,910,838 B2 *	12/2014	Palmer	A47B 23/002
D249,532 S *	9/1978	Cudahy, III	D19/113				108/43
				9,044,688 B1 *	6/2015	Janay	B65D 25/22
				9,468,290 B2 *	10/2016	Roth	A47B 19/06
				D977,025 S *	1/2023	Fan	D3/297
				2011/0056412 A1 *	3/2011	Grammer	A45C 9/00
							108/1
				2021/0170268 A1 *	6/2021	Malki	A63F 9/1044
				2023/0097737 A1 *	3/2023	Che	A63F 9/1044
							108/1

* cited by examiner

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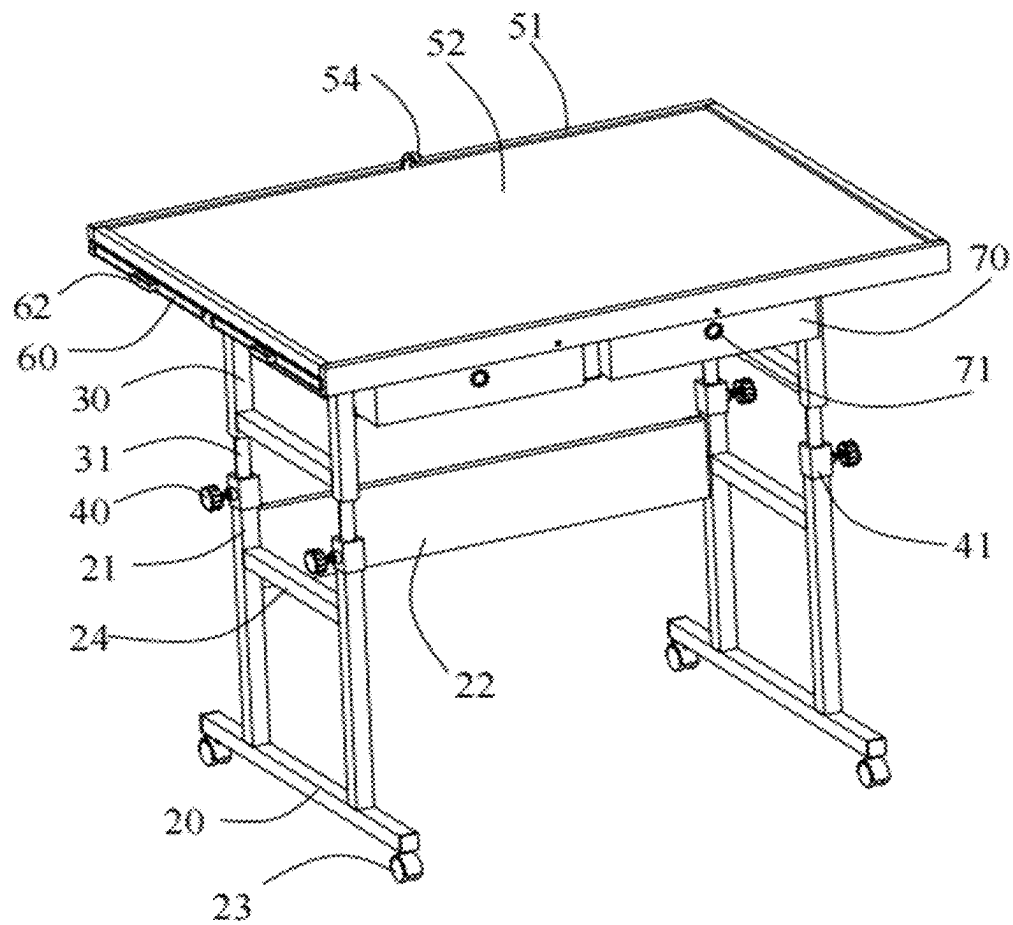


FIG. 1

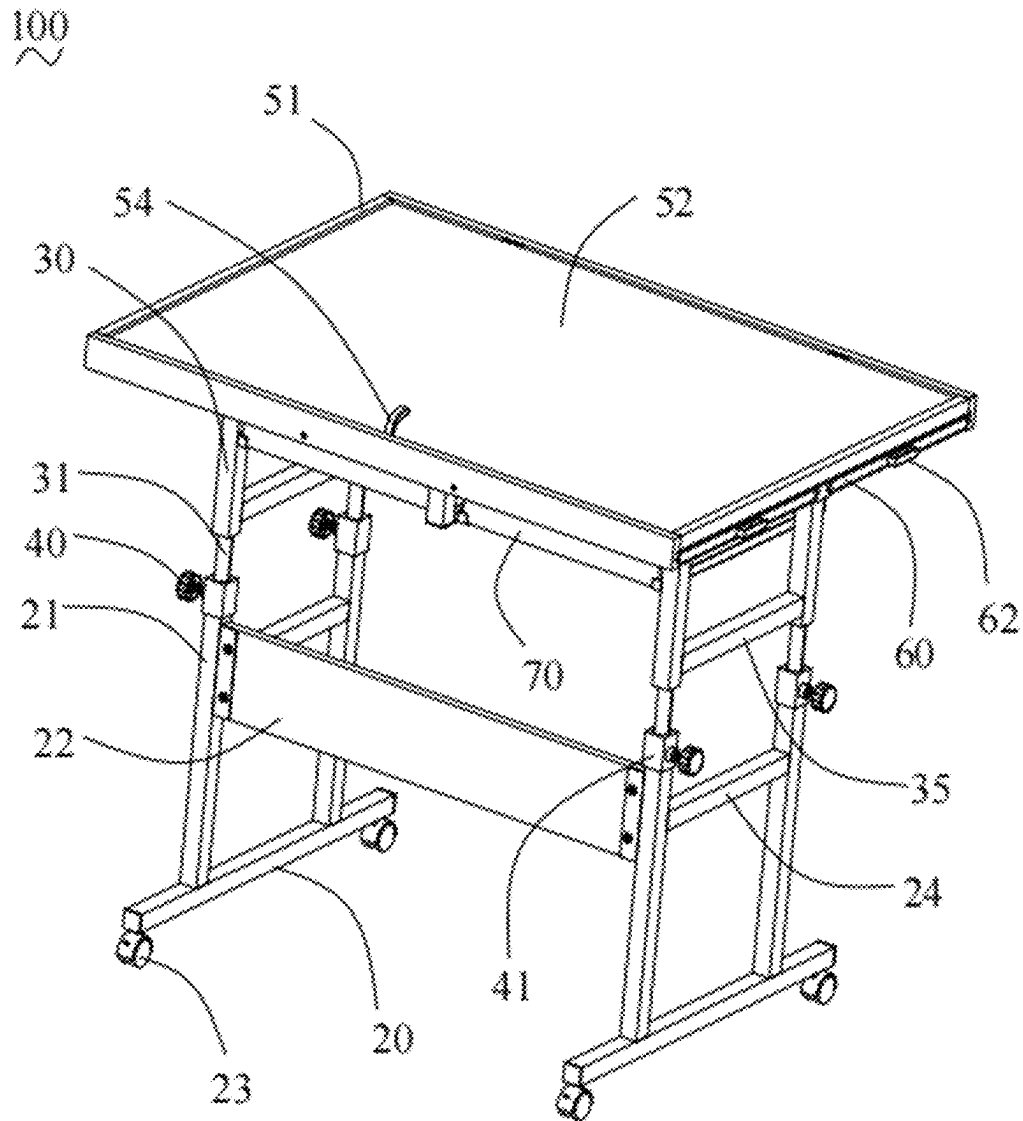


FIG. 2

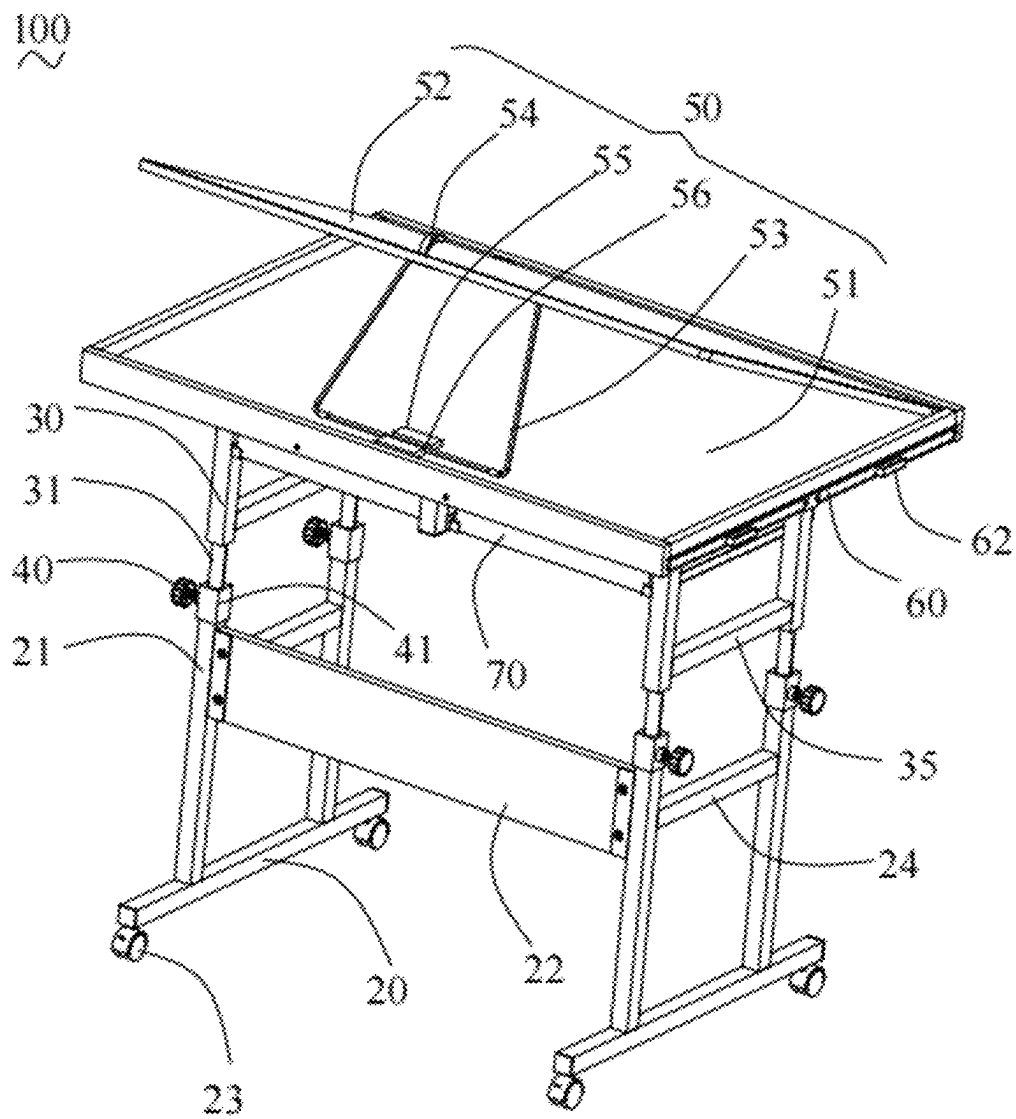


FIG. 3

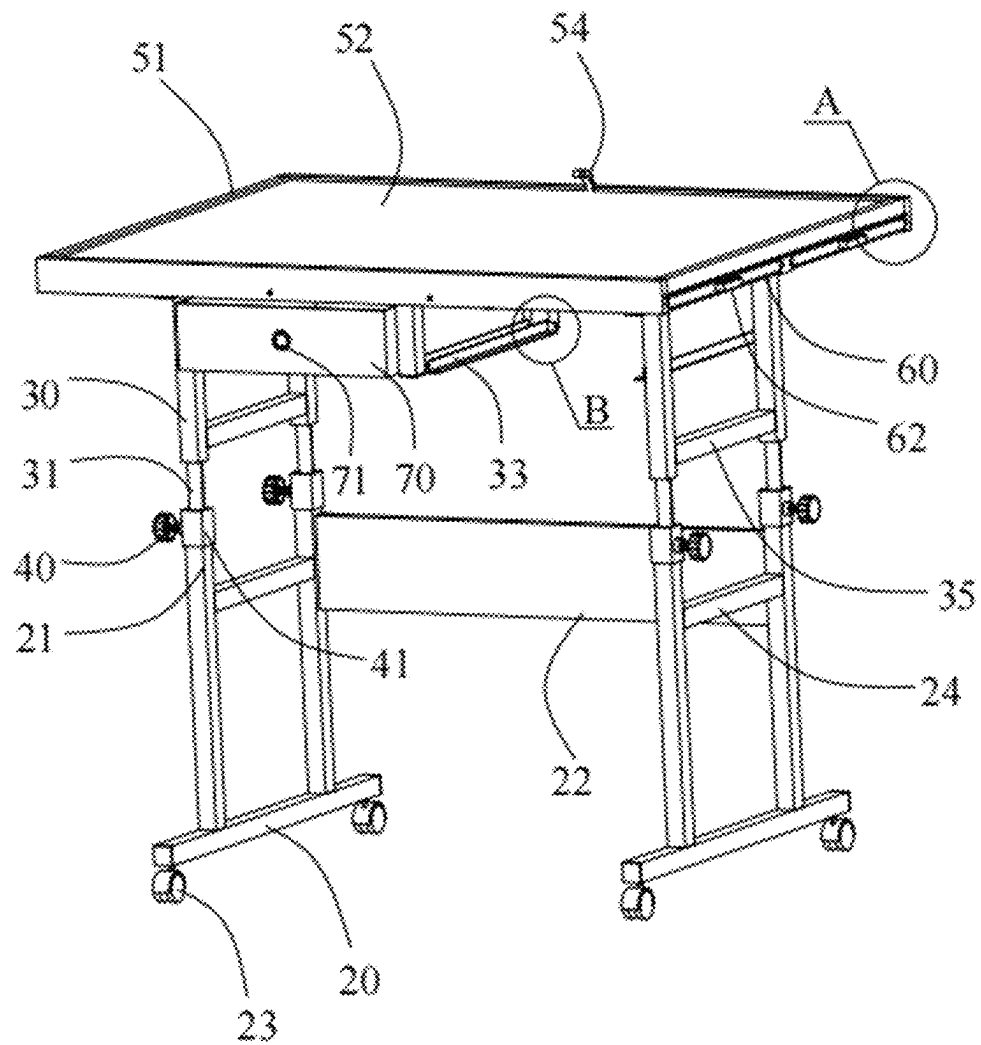


FIG. 4

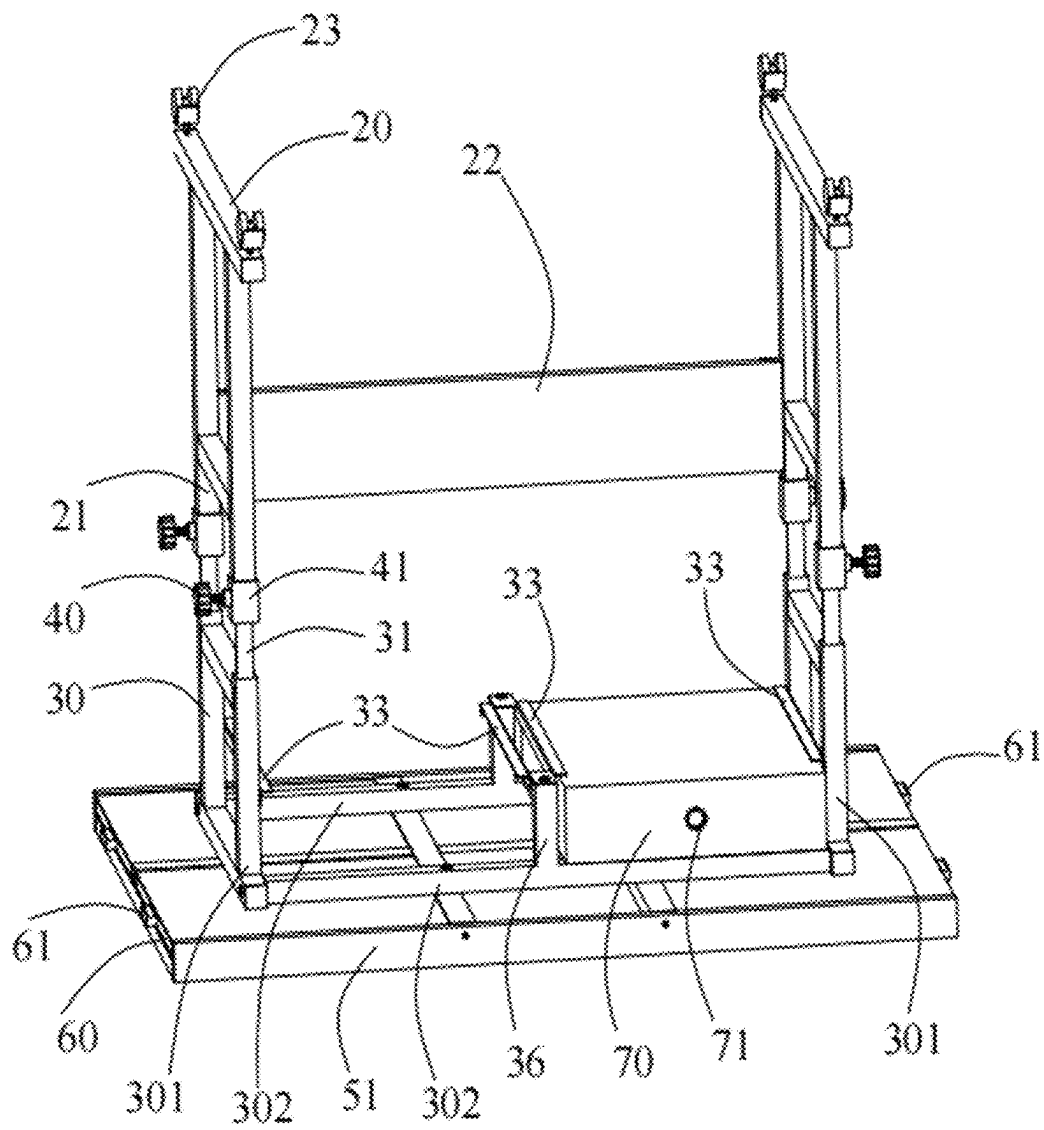


FIG. 5

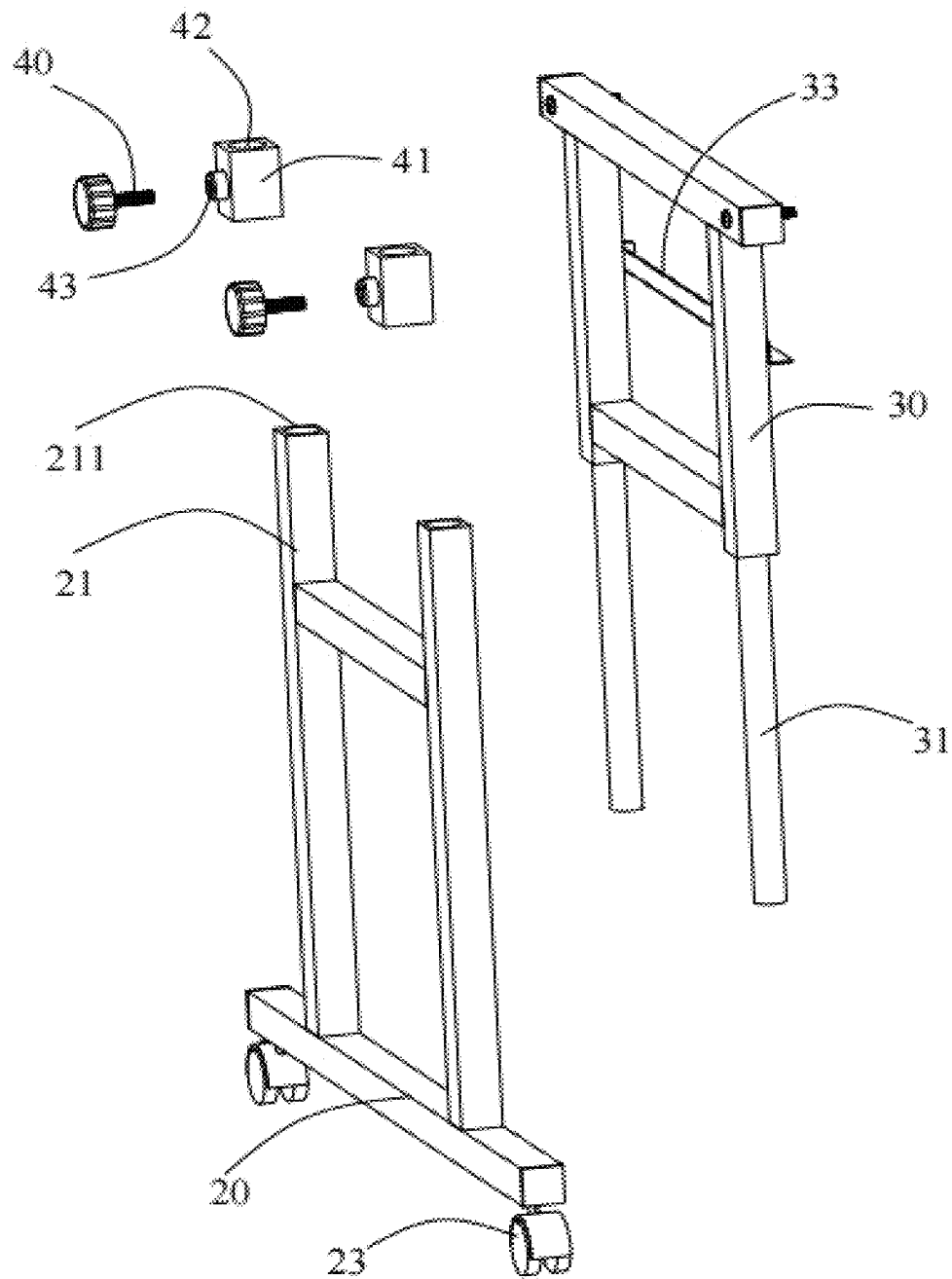


FIG. 6

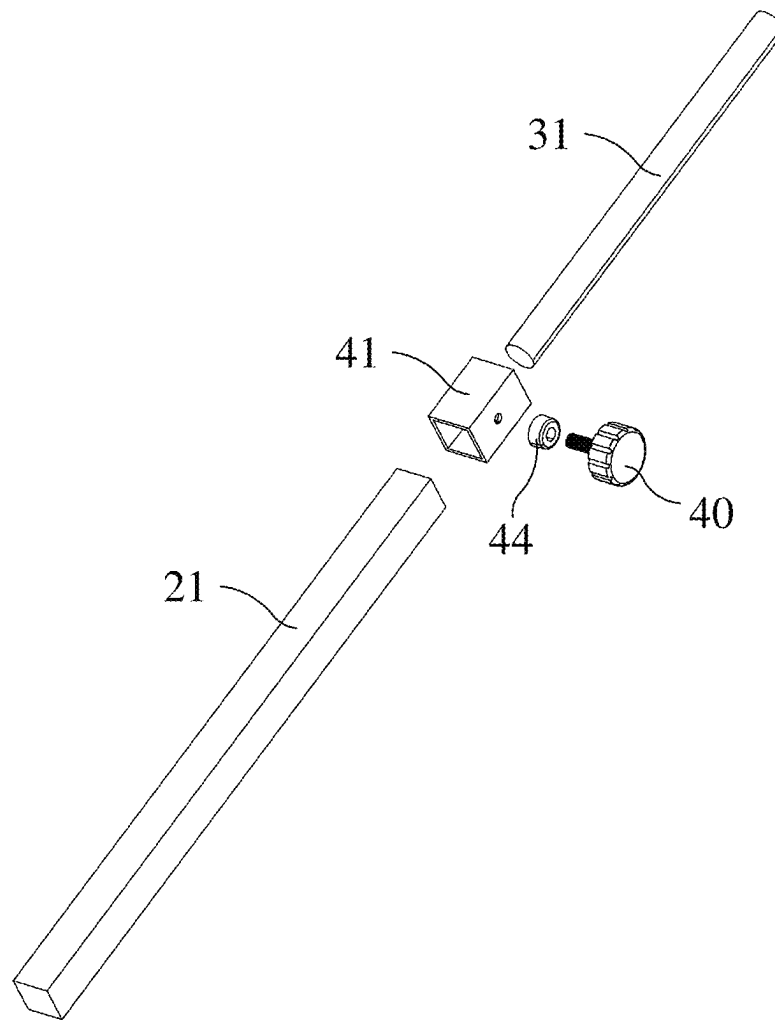


FIG. 7

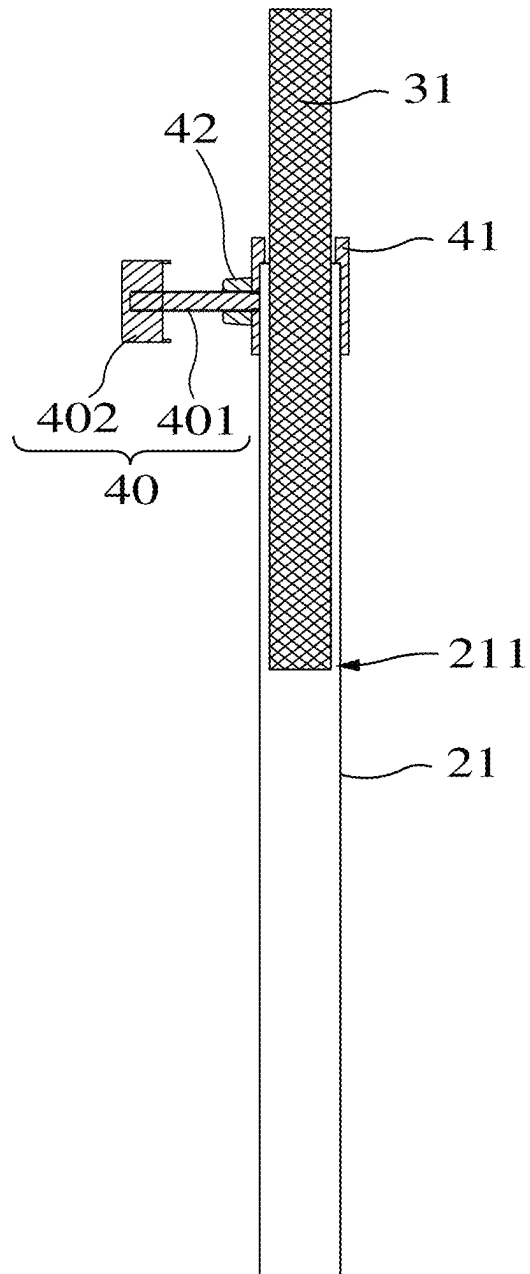


FIG. 8

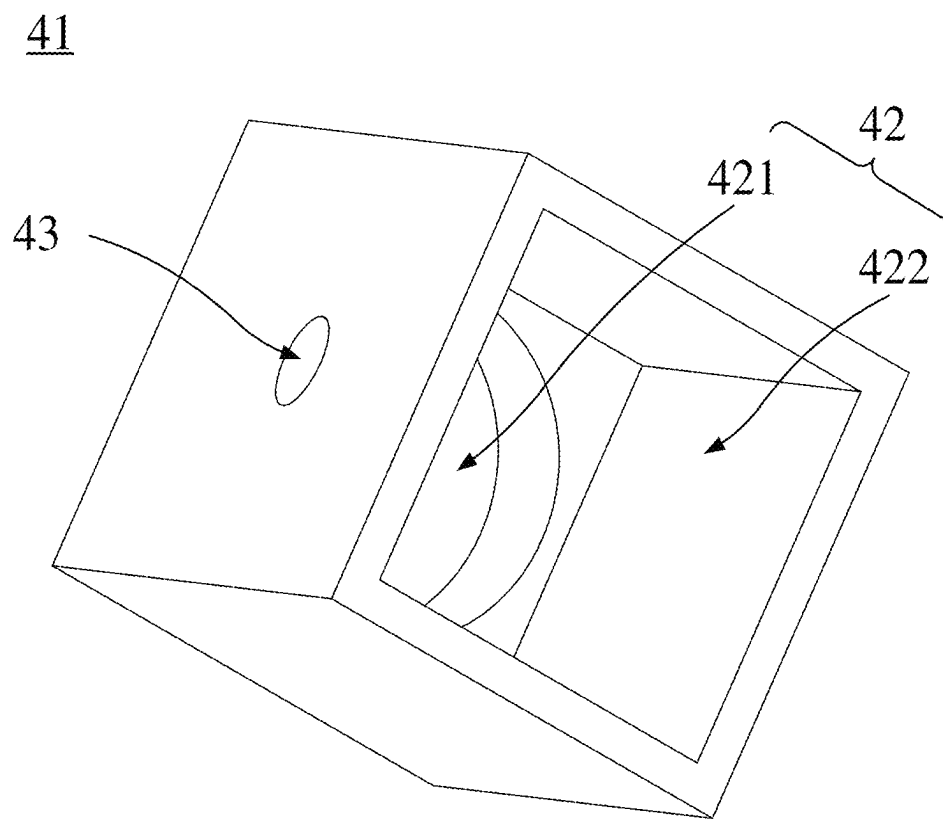


FIG. 9

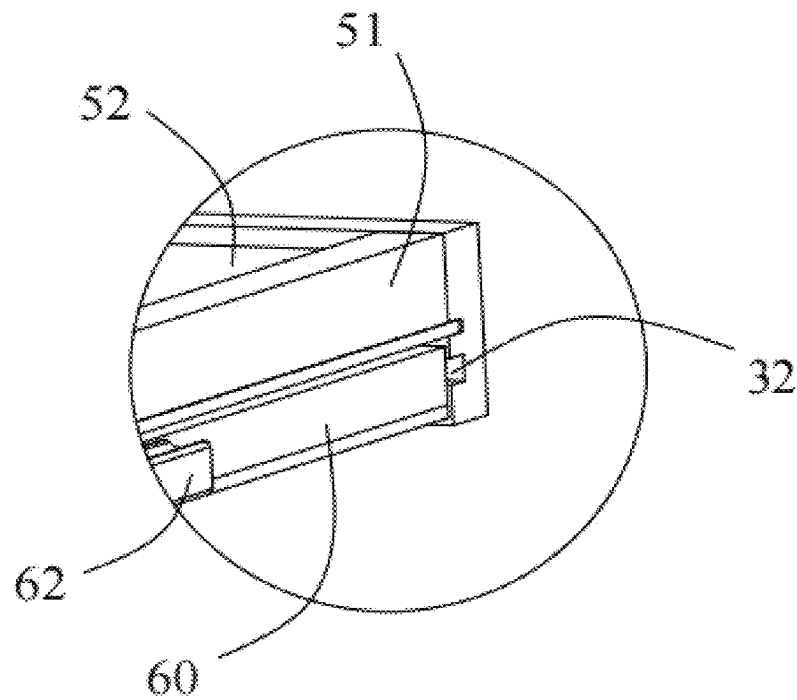


FIG. 10

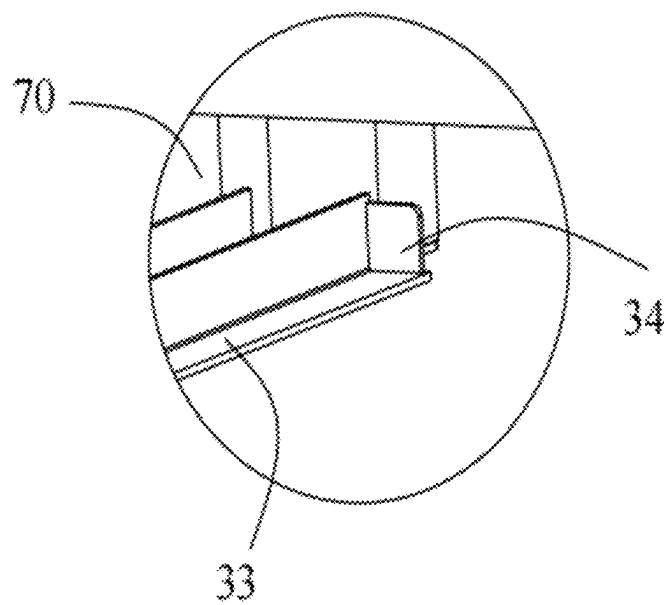


FIG. 11

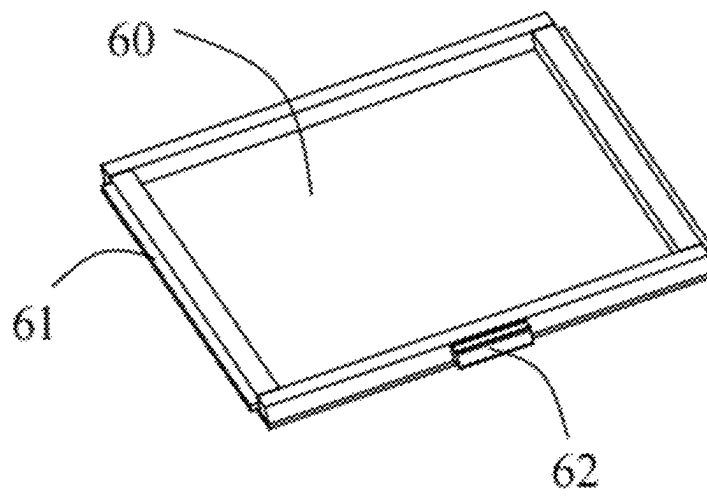


FIG. 12

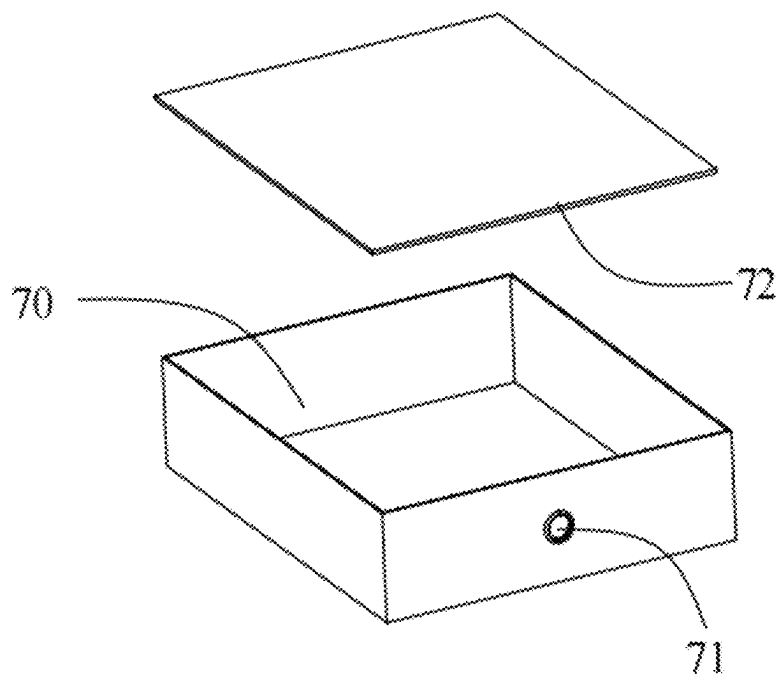


FIG. 13

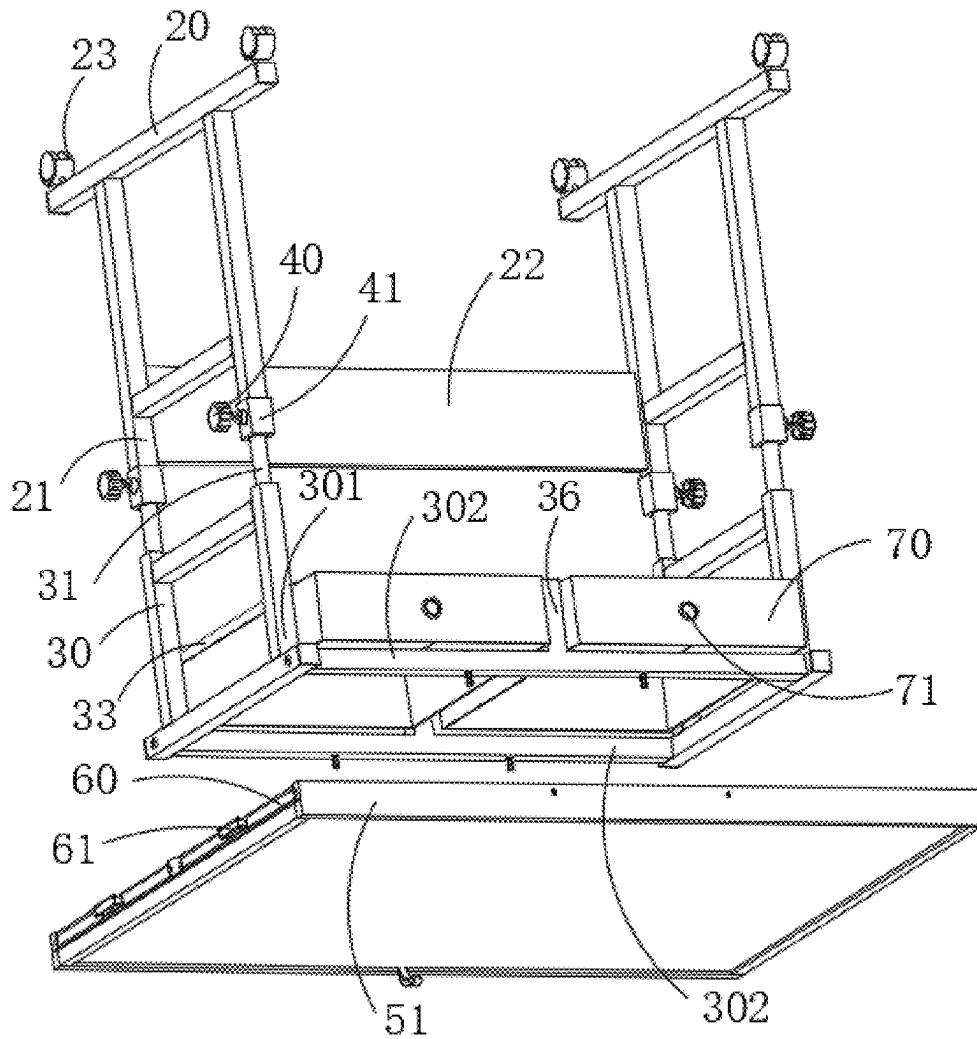


FIG. 14

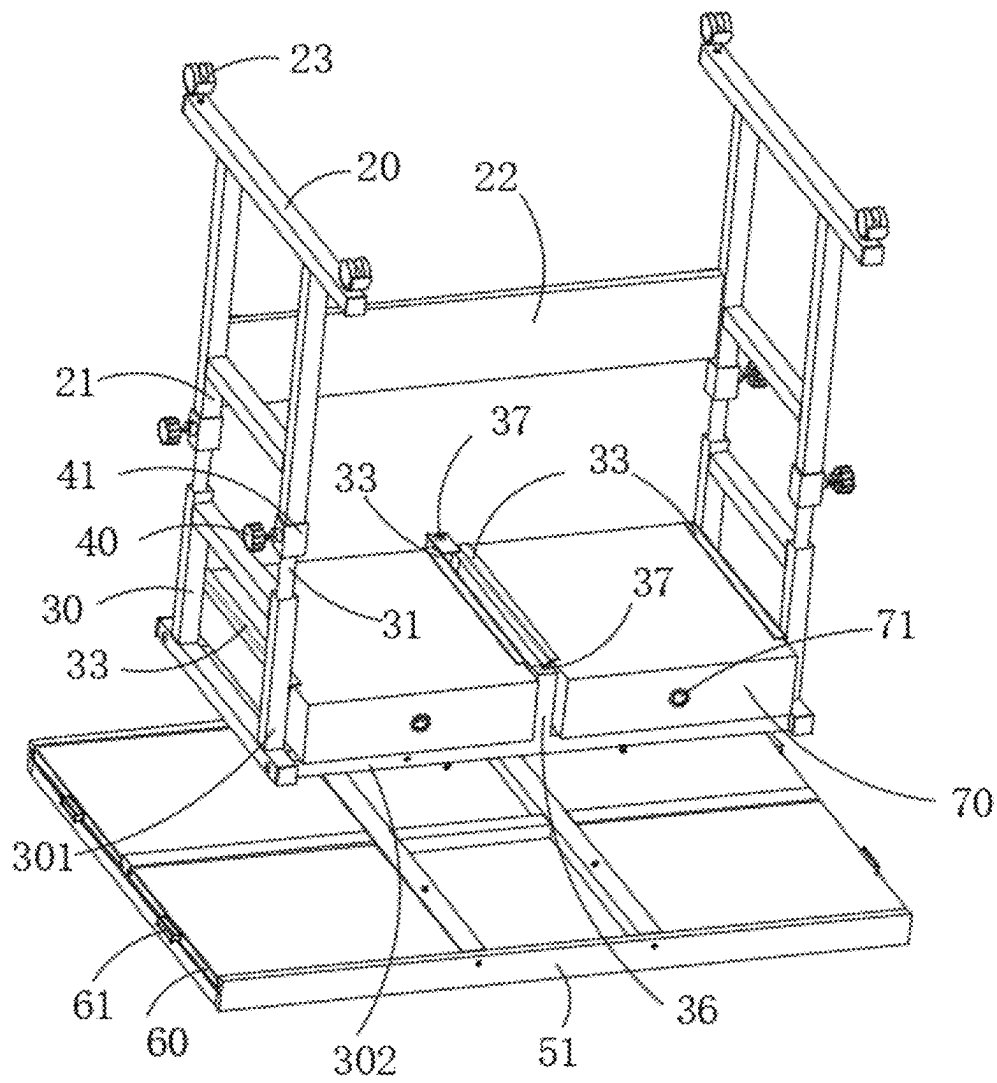


FIG. 15

JIGSAW PUZZLE TABLE**CROSS REFERENCE TO RELATED APPLICATIONS**

The present application is a continuation application of the U.S. application Ser. No. 18/740,863, filed on Jun. 12, 2024.

The U.S. application Ser. No. 18/740,863 is a continuation-in-part of the U.S. application Ser. No. 18/222,990, filed on Jul. 17, 2023.

The U.S. application Ser. No. 18/222,990 is a continuation-in-part application of the U.S. application Ser. No. 17/720,682, filed on Apr. 14, 2022, which claims the priority of the Chinese patent application No. CN202210103787.9, filed on Jan. 28, 2022.

The U.S. application Ser. No. 18/222,990 is a continuation-in-part application of the U.S. application Ser. No. 29/824,951, filed on Jan. 28, 2022.

The U.S. application Ser. No. 18/222,990 is a continuation-in-part application of the U.S. application Ser. No. 29/818,457, filed on Dec. 9, 2021.

The U.S. application Ser. No. 18/222,990 is a continuation-in-part of the U.S. application No. 29/859,220, filed on Nov. 8, 2022, which claims priority of the Chinese patent application No. CN202230717091.6, filed on Oct. 28, 2022.

Furthermore, the present application claims the foreign priority of the Chinese patent application No. 202420991164.4, filed on May 8, 2024, contents of which are incorporated herein by its entireties.

TECHNICAL FIELD

The present disclosure relates to the field of puzzle game accessories, and in particular to a jigsaw puzzle table.

BACKGROUND

The jigsaw puzzle, as a puzzle game, is essential in intellectual development for children. Due to the jigsaw puzzle being interesting, the jigsaw puzzle is also popular among adults. When playing the jigsaw puzzle, a player needs to carefully identify puzzle pieces to find similarities between the puzzle pieces to splice the puzzle pieces together. In order to enable the player to splice the puzzle pieces, the jigsaw puzzle table is released to the market. However, the jigsaw puzzle table in the art has a fixed height, and the height cannot be adjusted according to a height of the player. Therefore, the player whose height does not match the height of the jigsaw puzzle table may not use the jigsaw puzzle table comfortably, having a poor usage experience.

SUMMARY OF THE DISCLOSURE

The present disclosure provides a jigsaw puzzle table, including: a puzzle board; a board bracket, a lower support, and at least one bracket storage member. The puzzle board is mounted on a mounting end of the board bracket. The board bracket is mounted between the puzzle board and the lower support, the board bracket and the lower support are configured to support the puzzle board cooperatively. The at least one bracket storage member is configured to store puzzle pieces, and the at least one bracket storage member is supported by and movably mounted on the board bracket.

The present disclosure provides a jigsaw puzzle table, including: a puzzle board; a board bracket; a lower support;

and at least one bracket storage member. The puzzle board is mounted on a mounting end of the board bracket. The board bracket is mounted between the puzzle board and the lower support, the board bracket and the lower support are configured to support the puzzle board cooperatively. The at least one bracket storage member is configured to store puzzle pieces and disposed at a lower surface of the puzzle board, wherein the at least one bracket storage member is supported by and movably mounted on the board bracket.

The present disclosure provides a jigsaw puzzle table, including: a puzzle board; a board bracket; a lower support; at least one bracket storage member and at least one board storage member. The puzzle board is mounted on a mounting end of the board bracket. The board bracket is mounted between the puzzle board and the lower support, the board bracket and the lower support are configured to support the puzzle board cooperatively. At least one bracket storage member and at least one board storage member are configured to store puzzle pieces, wherein the at least one bracket storage member is supported by and movably mounted on the board bracket; the at least one bracket storage member is movably arranged below the at least one board storage member; and the at least one bracket storage member and the at least one board storage member are pullable in different direction.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic view of a jigsaw puzzle table according to an embodiment of the present disclosure.

FIG. 2 is a schematic view of the jigsaw puzzle table, being viewed from another viewing angle, according to an embodiment of the present disclosure.

FIG. 3 is a schematic view of the jigsaw puzzle table shown in FIG. 2 being in use.

FIG. 4 is a schematic view of the jigsaw puzzle table shown in FIG. 1 having a second storage portion omitted.

FIG. 5 is a schematic view of the jigsaw puzzle table shown in FIG. 4 being inverted.

FIG. 6 is an exploded view of a board bracket and a lower support of the jigsaw puzzle table shown in FIG. 1.

FIG. 7 is a schematic view of a support tube, a support rod, a positioning fastener, and a sleeve tube of the structure shown in FIG. 6.

FIG. 8 is a cross-sectional view of the structures shown in FIG. 7 in a length direction, after being assembled to each other.

FIG. 9 is a schematic view of the sleeve tube shown in FIG. 7.

FIG. 10 is an enlarged view of the portion A shown in FIG. 4.

FIG. 11 is an enlarged view of the portion B shown in FIG. 4.

FIG. 12 is a schematic view of a first storage portion of the jigsaw puzzle table shown in FIG. 1.

FIG. 13 is a schematic view of a second storage portion of the jigsaw puzzle table shown in FIG. 1.

FIG. 14 is an exploded view of the board bracket and the lower support shown in FIG. 1.

FIG. 15 is another exploded view of the board bracket and the lower support shown in FIG. 1.

DETAILED DESCRIPTION

In order to facilitate understanding of the present disclosure, the present disclosure will be fully described below by referring to the accompanying drawings. The accompanying

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drawings show preferred embodiments of the present disclosure. However, the present disclosure can be realized in various forms, which are not limited to the embodiments described herein. The embodiments are provided to enable the present disclosure to be understood more thoroughly and comprehensively.

It should be noted that when an element is “fixed” to another element, it means that the element is directly arranged on the another element, or an intermediate element is arranged therebetween. When an element is “connected” to another element, it means that the element is directly connected to the another element, or an intermediate element is arranged therebetween. Furthermore, the term “and/or” is merely a description of an associative relationship between associated objects and indicates that three relationships may exist. For example, A and/or B means that A exists alone, both A and B exist, and B exists alone.

Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by any ordinary skilled person in the art. Terms in the specification of the present disclosure are used only for the illustrative purposes only. For example, “in,” “out,” “left,” “right,” and similar expressions are used for illustrative purposes only, and are not intended to limit the present disclosure.

As shown in FIGS. 1 to 13, schematic views of the jigsaw puzzle table 100 are shown. The jigsaw puzzle table 100 includes a lower support 20, a board bracket 30, and a puzzle board 50. The puzzle board 50 is mounted at a mounting end of the board bracket 30. The board bracket 30 is mounted between the puzzle board 50 and the lower support 20. The lower support 20 and the board bracket 30 cooperatively support the puzzle board 50. The lower support 20 and the board bracket 30 are movably connected to each other adjust a support height on at least one side.

According to the puzzle table 100 of the present embodiment, the height of at least a portion of the puzzle table 50, which is supported by the lower support 20 and the board bracket 30 cooperatively, is adjusted by movably connecting the lower support 20 with the board bracket 30, such that the adjusted height meets demands of various users or meets various demands of a same user.

It should be understood that, in the present embodiment, the movable connection between the lower support 20 and the board bracket 30 to adjust the support height on at least one side of the puzzle board 50 means that the movable connection between the lower support 20 and the board bracket 30 adjusts the height of at least a portion of the puzzle board 50 which is supported by the lower support 20 and the board bracket 30.

In a first case, an overall of the lower support 20 and the board bracket 30 is movable. For example, the lower support 20 and the board bracket 30 can be displaced, such that an overall height of the puzzle board 50 supported by the lower support 20 and the board bracket 30 is adjusted. The entire puzzle board 50 may be adjusted from a first position to a second position. The first position and the second position are used to represent a change in position of the entire puzzle board 50.

In a second case, a first side of the lower support 20 is connected to a first side of the board bracket 30, and a second side of the lower support 20 is connected to a second side of the board bracket 30. The first side of the board bracket 30 supports a first portion of the puzzle board 50, and the second side of the board bracket 30 supports a second portion of the puzzle board 50. The first portion and the second portion of the puzzle board 50 may refer to

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portions near edges of the puzzle board 50 in a length direction of the puzzle board 50. A common side of the lower support 20 and the board bracket 30, i.e., either the first side or the second side thereof, is movable. For example, the first side thereof may be displaced, such that the height of the first portion of the puzzle board 50 supported by the first side of the lower support 20 and the first side of the board bracket 30 is adjusted, and the height of the second portion of the puzzle board 50 supported by the second side of the lower support 20 and the second side of the board bracket 30 is unchanged. In this way, the puzzle board 50 is changed from a first state to a second state. It should be understood that the first state is different from the second state. In the present disclosure, a support surface is defined, for example, the support surface refers to a surface on which the puzzle board 50 is placed. The support surface may be substantially parallel to a horizontal plane. Exemplarily, in the first state, the puzzle board 50 is substantially parallel to the support surface. In other words, the puzzle board 50 is substantially horizontally placed. In the second state, the puzzle board 50 is inclined with respect to the support surface, i.e., the puzzle board 50 is in an inclined state.

In a third case, at least one of the lower support 20 and the board bracket 30 has at least two sub-portions, and the puzzle board 50 has at least two sub-portions. One sub-portion of the at least two sub-portions of the lower support 20, one sub-portion of the at least two sub-portions of the board bracket 30, and one sub-portion of the at least two sub-portions of the puzzle board 50 are connected to each other correspondingly. At least one sub-portion of the lower support 20 is movably connected to the at least one sub-portion of the board bracket 30 to adjust the height of at least one sub-portion of the puzzle board 50. In an example, a first sub-portion of the lower support 20 is movably connected to a second sub-portion of the board bracket 30 to adjust a height of a third sub-portion of the puzzle board 50.

The movable connection between the lower support 20 and the board bracket 30 includes, but is not limited to, any one of sliding-sleeving connection, threaded connection, telescopic-driving connection, and splicing connection. The embodiments of the present disclosure will be illustrated by taking the sliding-sleeving connection between the lower support 20 and the board bracket 30 as an example. Exemplarily, a portion of one of the lower support 20 and the board bracket 30 can be inserted into the other one of the lower support 20 and the board bracket 30. In this way, sleeving connection between the lower support 20 and the board bracket 30 is achieved. In addition, the lower support 20 and the board bracket 30 can slide with respect to each other, such that the movable connection between the lower support 20 and the board bracket 30 is achieved. Therefore, the height of the puzzle board 50 supported by the lower support 20 and the board bracket 30 is adjusted.

It is to be understood that the sleeving connection between the lower support 20 and the board bracket 30 increases stability of the connection therebetween, the lower support 20 and the board bracket 30 are prevented from wobbling, such that stability of the jigsaw puzzle table 100 is improved.

To be noted that, at least a portion, such as at least one side (at least one of the first side and the second side), of the lower support 20 and the board bracket 30 of the present disclosure can be displaced. For example, the lower support 20 and the board bracket 30 are slidable relative to each other. In practice, when the height of the puzzle board 50 supported by the lower support 20 and the board bracket 30 reaches a height desired by the user, the lower support 20

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and the board bracket **30** are fastened to each other. Fastening between the lower support **20** and the board bracket **30** may be realized by the structures arranged on the lower support **20** and the board bracket **30** or by additional elements.

Exemplarily, the puzzle table **100** further includes a positioning fastener **40**, the positioning fastener **40** is connected to at least one of the lower support **20** and the board bracket **30**. The positioning fastener **40** is configured to secure the lower support **20** and the board bracket **30** to fix the height of puzzle board **50** supported by the lower support **20** and the board bracket **30**. The positioning fastener **40** is configured to release the fastening between the lower support **20** and the board bracket **30** to enable the lower support **20** and the board bracket **30** to at least partially move with respect to each other to adjust the support height of the puzzle board **50**.

In the present embodiment, the movable connection is the sliding-sleeving connection. The jigsaw puzzle table **100** includes: the lower support **20**; the board bracket **30** that is slidably and sleeveably connected to the lower support **20**; the positioning fastener **40** that is mounted on the lower support **20** and is opposite to the board bracket **30**; and the puzzle board **50** mounted on the board bracket **30**. The board bracket **30** may be lifted up or lowered down along the lower support **20**.

Exemplarily, the lower support **20** has four support tubes **21**. The board bracket **30** has four support rods **31**. One end of each support rod **31** is sleeved inside a respective one support tube **21**. In this way, the board bracket **30** and the lower support **20** are guided to be assembled to each other at four points, such that stability of the support is improved. Further, the lower support **20** is arranged with a reinforcing plate **22**, and the reinforcing plate **22** is configured to improve overall structural strength of the lower support **20**.

As shown in FIGS. **6** to **9**, in the present embodiment, the jigsaw puzzle table **100** further includes a sleeve tube **41** that sleeves the support tube **21** and the support rod **31**. The positioning fastener **40** is connected to the sleeve tube. The sleeve tube **41** is configured to adjust a relative positional relationship, or a connection relationship or a distance relationship, between the support tube **21** and the support rod **31**, such that the height that the support tube **21** and the support rod **31** support the puzzle board **50** is adjusted.

Exemplarily, after the sleeve tube **41** sleeves the support tube **21** and the support rod **31**, the sleeve tube **41** is fastened to the support tube **21** and/or the support rod **31** by the positioning fastener **40**, such that the support tube **21** and the support rod **31** are positioned. It should be understood that the positioning fastener **40** can disconnect the sleeve tube **41** from the support tube **21** and/or the support rod **31**, such that the relative positional relationship between both the support tube **21** and the support rod **31** is adjusted.

Exemplarily, the positioning fastener **40** includes a fastening portion **401** and an operating portion **402**. The fastening portion **401** is connected to the operating portion **402**. The fastening portion **401** protrudes from an end face of the operating portion **402**. The fastening portion **401** is connected to at least the sleeve tube **41**. Exemplarily, the sleeve tube **41** defines a fastening hole **43**. The fastening portion **401** can be received in and detached from the fastening hole **43**.

In order to increase stability of connection between the fastening portion **401** and the sleeve tube **41**. A reinforcing member **44** is arranged on an outer surface of the sleeve tube **41**. The reinforcing member **44** is arranged around the fastening hole **43**. The fastening portion **401** passes through

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the reinforcing member **44** to be connected to the sleeve tube **41**. The reinforcing member **44** can be connected to the sleeve tube **41** by bonding or welding, or the reinforcing member **44** and the sleeve tube **41** are integrally molded to form a one-piece structure. The reinforcing member **44** increases a connection area between the fastening portion **401** and sleeve tube **41**, and therefore, stability of the connection therebetween is improved.

The connection between the fastening portion **401**, the sleeve tube **41**, and the reinforcement member **44** includes, but is not limited to, threaded connection and snap connection. The embodiments of the present disclosure will be illustrated by taking the threaded connection between the fastening portion **401**, the sleeve tube **41**, and the reinforcing member **44** as an example. In this way, connection and separation therebetween can be achieved easily.

The operating portion **402** is operated by a finger of the user or by a tool. Exemplarily, a non-slip element or an operating element for operation by the tool is arranged around a circumference of the operating portion **402**. To be noted that the structure of the positioning fastener **40** and the connection between the positioning fastener **40** and the sleeve tube **41** are not limited the present disclosure.

Exemplarily, the support rod **31** and the support tube **21** are connected to each other in a sleeved manner. For example, the support rod **31** can be inserted into the support tube **21** and is movable, such as slidable, inside the support tube **21**. Exemplarily, the support tube **21** defines a receiving channel **211**. The support rod **31** is received in the receiving channel **211** and is movable in the receiving channel **211**. It should be understood that the support rod **31** passes through the sleeve tube **41** firstly, and subsequently, inserted into the receiving channel **211**. In the present embodiment, the sleeve tube **41** is hollow, ensuring that the support rod **31** passes through the sleeve tube **41**. The sleeve tube **41** defines a tube channel **42**.

To be noted that a depth of the receiving channel **211** may be determined according to the actual needs, which will not be limited herein.

Exemplarily, the sleeve tube **41** defines a first channel **421** and a second channel **422** communicated with the first channel **421**. A shape of the first channel **421** is adapted to a shape of an outer surface of the support rod **31**, and a shape of the second channel **422** is adapted to a shape of an outer surface of the support tube **21**.

It should be understood that the first channel **421** and the second channel **422** enables the sleeve tube **41** to be hollow. A size of the second channel **422** is not smaller than a size of the first channel **421**. The second channel **422** allows the support rod **31** to pass through. Exemplarily, the size of the second channel **422** is greater than the size of the first channel **421**. An outer periphery of the support tube **21** is larger than the size of the first channel **421**. After the sleeve tube **41** sleeves the support tube **21**, the support tube **21** is restricted in the second channel **422** and is prevented from extending into the first channel **421**.

Exemplarily, a cross section of the support rod **31** is circular, a cross section of the support tube **21** is square, the first channel **421** is a circular channel, and the second channel **422** is a square channel. To be noted that, in the present embodiment, the shape of the support rod **31** and the shape of the support tube **21** are not limited by the present disclosure. The shapes herein are exemplary only and do not limit the shapes thereof.

Exemplarily, the support tube **21** is quadrangular, the support rod **31** is cylindrical, and the receiving channel **211** is a circular channel. The tube channel **42** includes a circular

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channel and a quadrangular channel communicated with the circular channel. In this way, the sleeve tube **41** can sleeve the support tube **21** through the quadrangular channel, and the sleeve tube **41** and the support tube **21** may not rotate relative to each other. The support rod **31** may be inserted into the receiving channel **211** by passing through the circular channel.

In the present embodiment, the fastening hole **43** is communicated to the first channel **421** and/or the second channel **422**, such that it is ensured that the fastening portion **401** passes through the fastening hole **43** firstly, and subsequently, passes through the first channel **421** and/or the second channel **422** to abut against the support rod **31** and/or the support tube **21**.

In other embodiments, in order to increase the fastening function of the positioning fastener **40**, the support rod **31** and/or the support tube **21** may define a plurality of holes or slots in which the positioning fastener **40** can be received. The positioning fastener **40** may be inserted different holes in the support rod **31** and/or the support tube **21** to adjust the relative positions between the support rod **31** and the support tube **21**.

By arranging the sleeve tube **41**, fastening support strength of a fastening bolt can be improved, such that a damage caused by insufficient support strength between the fastening bolt and the support tube **21**, due to the support tube **21** being excessively thin, can be avoided. When in use, when a height of the board bracket **30** relative to the lower support **20** needs to be adjusted, the fastening bolt is firstly threaded outwardly by a certain distance, and then the support rod **31** is moved upwardly or downwardly by a predetermined distance inside the receiving channel **211**. Subsequently, the fastening bolt is moved inwardly by a certain distance to abut against an outer wall of the support rod **31** and is further threaded to fasten the support rod **31** to the support tube **21**. In an embodiment, in order to enable the entire jigsaw puzzle table **100** to be moved, four universal wheels **23** having self-locking switches are mounted at a bottom of the lower support **20**.

Exemplarily, in an embodiment, the lower support **20** is arranged with a first crossbeam **24**. The board bracket **30** is arranged with a second crossbeam **35**. By arranging first crossbeams **24** and second crossbeams **35** at opposite sides of the jigsaw puzzle table **100**, overall strength and stability of the jigsaw puzzle table **100** is improved, and the jigsaw puzzle table **100** may not be collapsed after being used for a long period of time. One first crossbeam **24** connects at least two support tubes **21** to increase strength of the lower support **20**. One second crossbeam **35** connects at least two support rods **31** of the board bracket **30** to increase strength of the board bracket **30**.

Exemplarily, the first crossbeam **24** is located at a side of the lower support **20** near the board bracket **30**, and the second crossbeam **35** is located at a side of the board bracket **30** near the lower support **20**. In this way, strength of a position where the lower support **20** is connected to the board bracket **30** is improved. In addition, during the lower support **20** and the board bracket **30** are movably connected to each other, the first crossbeam **24** and the second crossbeam **35** increase the stability of the connection therebetween and limit, to some extent, a moving range of the lower support **20** and the board bracket **30**, such that the lower support **20** and the board bracket **30** are movable within a predetermined range.

Exemplarily, in the present embodiment, the positioning fastener **40** is a fastening bolt, and the lower support **20** defines a thread hole (not shown) correspondingly. When in

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use, the fastening bolt is threaded to move inwardly or outwardly. When the fastening bolt is threaded to move inwardly to abut against the outer wall of the board bracket **30**, the fastening is achieved. When the fastening bolt is threaded to move outwardly to leave the board bracket **30**, the fastening is released. In this case, the board bracket **30** is moved up and down to adjust the support height.

In another embodiment, the positioning fastener **40** is a pin. In this case, the support tube **21** of the lower support **20** defines a pair of insertion holes. The support rod **31** of the board bracket **30** defines a plurality of pairs of insertion holes, and the plurality of pairs of insertion holes are equidistantly spaced apart from each other. The pin passes through the pair of insertion holes in the support tube **21** and one pairs of insertion holes in the support rod **31**, such that the height is adjusted with multiple gradients.

In other embodiments, the positioning fastener **40** may be configured as other structures, as long as the lower support **20** and the board bracket **30** can be fastened with and detached apart from each other. the present disclosure does not limit detailed structure of the positioning fastener **40**.

When the movable connection is the threaded connection, the lower support **20** is arranged with an externally-threaded tube, the board bracket **30** is arranged with an externally-threaded tube, and adjacent externally-threaded tubes are connected to each other by an internally-threaded tube sleeve. In this way, when the support height is adjusted, a relative height difference between the lower support **20** and the board bracket **30** can be changed by threading the internally-threaded tube sleeve, such that the support height formed by the lower support **20** and the board bracket **30** can be adjusted. When the movable connection is the telescopic-driving connection, a telescopic drive cylinder is mounted on the lower support **20**, and a telescopic rod on the telescopic drive cylinder is connected to the board bracket **30**. The telescopic drive cylinder drives the board bracket **30** to move up and down to change the support height. When the movable connection is splicing connection, each of the lower support **20** and the board bracket **30** is formed by splicing a plurality of blocks. Each block defines a rod hole in which a rod is inserted. In this way, the support height is adjusted by the lower support **20** and rod insertion.

In the present embodiment, the lower support **20** may be a one-piece frame, and the board bracket **30** may be a one-piece frame. In this way, heights of two sides of the board bracket **30** can be adjusted synchronously. In other embodiments, the lower support **20** may be formed by a plurality of support assemblies assembled to each other, and the board bracket **30** may be formed by a plurality of bracket assemblies assembled to each other. The plurality of support assemblies and the plurality of bracket assemblies may be connected in one-to-one correspondence to each other to form a plurality of support surfaces at a plurality of sides. In this way, the support height of one side or support heights of the plurality of sides can be adjusted based on demands. When there is a difference in heights of different support surfaces, the puzzle board **50** is tilted at an angle relative to the horizontal plane.

Exemplarily, the puzzle board **50** includes a puzzle carrying board **52**, a support board **51**, and a movable support member **53**. The puzzle carrying board **52** is arranged above the support board **51**. The movable support member **53** is disposed between the puzzle carrying board **52** and the support board **51**. The puzzle carrying board **52** is switchable, by the movable support member **53**, between a state in which the puzzle carrying board **52** is placed flat on the

support plate 51 and a state in which the puzzle carrying board 52 is tilted to have an angle with respect to the support plate 51.

One end of the puzzle carrying board 52 is hinged to one end of the support board 51, forming a hinged portion. The movable support member 53 is disposed below a portion of the puzzle carrying board 52 away from the hinged portion. Specifically, one end of the movable support member 53 is hingedly connected to the puzzle carrying board 52 or the support board 51.

Exemplarily, as shown in FIG. 3, in the present embodiment, one end of the movable support member 53 is hinged to the puzzle carrying board 52. A carrying handle 54 is mounted on a top surface of the puzzle carrying board 52. The carrying handle 54 is disposed on an end of the puzzle carrying board 52 away from the hinged portion. Further, the support plate 51 is arranged with two oppositely disposed limiting strips 55. The two limiting strips 55 cooperatively define a limiting slot 56. The movable support member 53 is adapted to the limiting slot 56. When in use, the carrying handle 54 is held by a hand of the user, and the puzzle carrying board 52 is pulled upwardly from the support plate 51. Subsequently, the movable support member 53 at the bottom of the puzzle carrying board 52 may swing at a certain angle to enable an end portion thereof to be received in the limiting slot 56. In this way, the puzzle carrying board 52 is tilted at a certain angle, and the user may play with jigsaw puzzle pieces.

In other embodiments, an end of the movable support member 53 is hinged with the support plate 51. Two oppositely disposed limiting strips are mounted on a bottom surface of the puzzle carrying board 52. When in use, the other end of the movable support member 53 is received between the two limiting strips and is limited by the two limiting strips.

In other embodiments, the puzzle carrying board 52 abuts against the support plate 51. In this case, the movable support member 53 may be an inclined support stage, and a height thereof is adjustable. In this way, the movable support member 53 may be placed on the support plate 51 to abut against the bottom surface of the puzzle carrying board 52, such that the puzzle carrying board 52 is inclined with respect to the support plate 51.

Exemplarily, the jigsaw puzzle table 100 further includes at least one first storage member 60 and/or at least one second storage member 70. For example, the at least one first storage member 60 may be referred to as at least one board storage member 60, as the first storage member 60 is movably mounted inside the puzzle board 50, more specifically, mounted with the support plate 51 and below a puzzle surface of the puzzle board 50. The first storage member 60 is movable relative to the puzzle board 50 to be opened or closed. The second storage member 70 may be referred to as a bracket storage member, as the second storage member 70 is movably mounted below the puzzle surface of the puzzle board 50 and is located out of the puzzle board 50. More specifically, the second storage member 70 is mounted in and supported by the board bracket 30. The second storage member 70 is movable relative to the puzzle board 50 to be opened or closed. Further, a storage cavity size of the first storage member 60 is different from a storage cavity size of the second storage member 70. The first storage member 60 and the second storage member 70 are mounted at different sides of the puzzle board 50. In the present embodiment, a depth of the storage cavity of the second storage member 70 is greater, and therefore, puzzle pieces in a larger size and a larger thickness can be stored in the second storage member

70. In the present embodiment, the first storage member 60 and the second storage member 70 are disposed below the puzzle board 50. In other embodiments, the first storage member 60, the second storage member 70 may be mounted on the puzzle board 50 when a thickness of the puzzle board 50 is large enough, and in this case, the first storage member 60 and the second storage member 70 are located in a cavity below the puzzle surface of the puzzle board 50.

The puzzle surface of the puzzle board 50 refers to a surface away from the lower support 20, or a surface on which the user splices the puzzle pieces.

Exemplarily, the first storage member 60 is located above the second storage member 70. The first storage member 60 is mounted on a side (a short edge) of the puzzle board 50, and the second storage member 70 is mounted on a front (a long edge) of the puzzle board 50. Specifically, a height of the first storage member 60 along a height direction of the puzzle board 50 is greater than a height of the second storage member 70. In this way, the first storage member 60 and the second storage member 70 are disposed at different planes and do not collide each other when being moved. Specifically, to avoid the collision, a bottom surface of the first storage member 60 is higher than a top surface of the second storage member 70.

It should also be noted that, in the present embodiment, the depth of the second storage member 70 disposed on the front of the puzzle board 50 is configured to be greater than the depth of the first storage member 60 disposed on the side of the puzzle board 50. In this way, collision between the second storage member 70 and the first storage member 60, when being moved, is reduced, and in addition, items can be stored therein. It is understood that a length of the front of the puzzle board 50 is greater than a length of the side of the puzzle board 50. In other words, the length of the long edge of the puzzle board 50 is greater than the length of the short edge of the puzzle board 50. In practice, a frequency that the user faces the front of the puzzle board 50 to splice puzzle pieces is higher a frequency that the user faces the size of the puzzle board 50 to splice puzzle pieces. Furthermore, since the length of the side is smaller than the length of the front, more users may face the front of the puzzle board 50 to splice puzzle pieces. Therefore, by considering various factors, the depth of the second storage member 70 disposed at the front of the puzzle board 50 is greater than the depth of the first storage member 60 disposed at the size of the puzzle board 50. Therefore, the user may store or take out items conveniently. More specifically, the first storage member 60 disposed on the side of the puzzle board 50 means that the first storage member 60 may be pulled out of the puzzle board 50 from a short edge of the puzzle board 50, and the second storage member 70 disposed on the front of the puzzle board 50 means that the second storage member 70 may be pulled out of the puzzle board 50 from a longer edge of the puzzle board 50.

Exemplarily, the board bracket 30 defines at least one set of sliding slots 32 and/or at least one set of support retaining edges 33. The first storage member 60 is slidably connected to the sliding slots 32, and the second storage member 70 is slidably connected to the support retaining edges 33. In this way, the first storage member 60 may slide along the sliding slots 32, and the second storage member 70 may slide along the support retaining edges 33. Therefore, the sliding is guided and stable.

Exemplarily, as shown in FIGS. 1 to 5, 10 and 12, the jigsaw puzzle table 100 further includes at least one first storage member 60 mounted on a side end of the board bracket 30. The board bracket 30 defines at least one set of

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sliding slots 32, and the first storage member 60 is arranged with slide rails 61 adapted to the sliding slots 32, and the slide rails 61 are mounted on two sides of the first storage member 60. In this way, the first storage member 60 may slide smoothly along the sliding slots 32. In order to facilitate the first storage member 60 to be pulled out and received in the board bracket, the first storage member 60 is arranged with a pulling block 62. In the present embodiment, four first storage members 60 are arranged, and four sets of sliding slots 32 are defined. Two of the four first storage members 60 are located on one side of the board bracket 30, and the other two of the four first storage members 60 are located on the other side of the board bracket 30. In this way, four pullable storage regions are formed, puzzle pieces in smaller sizes and smaller thicknesses can be stored.

Exemplarily, as shown in FIGS. 4, 5, 11, and 13, the jigsaw puzzle table 100 further includes at least one second storage member 70 mounted on the board bracket 30. The second storage member 70 is disposed under the puzzle board 50. The board bracket 30 is arranged with at least one set of support retaining edges 33. Two side edges of the second storage member 70 abut against the support retaining edges 33, respectively. The second storage member 70 can be pulled under the support and the limitation of the support retaining edges 33. Further, an end of the second storage part 70 defines a pulling hole 71, and a limiting block 34 is arranged on each support retaining edge 33. The limiting block 34 is located on an end of the support retaining edge 33 away from the pulling hole 71. In this way, when in use, the finger may apply a pulling force to the pulling hole 71, which is quite convenient. The limiting block 34 ensures that the second storage member 70 is not be pushed in excessively far, and therefore, the second storage member 70 is prevented from sliding off from the support retaining edge 33. In the present embodiment, two second storage members 70 are arranged, and two sets of support retaining edges 33 are arranged. In this way, two pullable storage regions are formed, and puzzle pieces of larger size and larger thicknesses can be stored conveniently. In the present embodiment, a pair of bracket rods 36 are mounted at a bottom of the board bracket 30. The bracket rods 36 are configured to support the support retaining edges 33. Ends of the support retaining edges 33 are connected to the bracket rods 36, corresponding.

As shown in FIG. 5, the board bracket 30 includes two oppositely disposed side end portions 301 and two crossbar portions 302. The above-mentioned support rods 31 are disposed at bottoms of the side end portions 301 extending downwardly. End portions of the two crossbar portions 302 are respectively connected to the corresponding side end portions 301. In the present embodiment, the bracket rods 36 extend vertically downwardly from bottoms of the crossbar portions 302. The bracket rods 36 and the crossbar portions 302 may be connected to each other by bonding or welding, or the bracket rods 36 and the crossbar portions 302 may be integrally molded to form a one-piece structure, which will not be limited herein. Each bracket rod 36 is located at a middle between the two side end portions 301. In this way, forces are more uniformly distributed, and the bracket rods 36 provide a better lifting and supporting effect. Two support retaining edges 33 are mounted between the two bracket rods 36 and are opposite to each other. Each side end portion 301 is arranged with one support retaining edge 33 at a side near the bracket rods 36. In this way, a side of the second storage member 70 is disposed on the support retaining edge 33 arranged on the bracket rods 36, and the other side of the

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second storage member 70 is disposed on the support retaining edge 33 arranged on the side end portion.

FIG. 14 and FIG. 15 further show exploded view of the board bracket and the lower support. In particular, the support retaining edge 33 is connected to the bracket rod 36 via a screw 37. For example, an end of the support retaining edge 33 near the pulling hole 71 of the second storage member 70 is connected to the bracket rod 36 through a screw 37, and the other end of the support retaining edge 33 away from the pulling hole 71 of the second storage member 70 is connected to the bracket rod 36 through another screw 37.

In another embodiment, a second storage cover 72 is further arranged for the second storage member 70. The second storage cover 72 is configured to cover the second storage member 70 when the second storage member 70 is removed from the board bracket 30 and used as a storage box.

According to the above-described jigsaw puzzle table 100, since the board bracket 30 is mounted between the puzzle board 50 and the lower support 20. The lower support 20 and the board bracket 30 cooperatively support the puzzle board 50. The lower support 20 and the board bracket 30 are movably connected to each other so as to adjust the support height at at least one side of two sides. When in use, when the user feels uncomfortable and needs to adjust the height, the player only needs to adjust the positional relationship between the lower support 20 and the board bracket 30 at one or more sides, such that the support height formed between the lower support 20 and the board bracket 30 is adjusted, and a comfortable height for the user is reached. The height of the jigsaw puzzle table 100 can be adjusted according to the height of the user, and a better usage experience is provided.

Various technical features of the above-described embodiments can be combined arbitrarily. In order to make the description concise, not all possible combinations of the various technical features are described. However, as long as no contradiction occurs, any combination thereof should be included in the scope of the present disclosure.

The above-described embodiments show only several embodiments of the present disclosure, which are described in a more specific and detailed manner, but shall not be interpreted as a limitation of the scope of the present disclosure. To be noted that, any ordinary skilled person in the art may perform various deformations and improvements without departing from the concept of the present disclosure, all of which shall fall within the scope of the present disclosure. Therefore, the scope of the present disclosure shall be subject to the appended claims.

What is claimed is:

1. A jigsaw puzzle table, comprising:

- a puzzle board;
- a board bracket, wherein the puzzle board is mounted on a mounting end of the board bracket; and
- a lower support, wherein the board bracket is mounted between the puzzle board and the lower support, the board bracket and the lower support are configured to support the puzzle board cooperatively;
- at least one bracket storage member, configured for storage, wherein the at least one bracket storage member is supported by and movably mounted on the board bracket;
- wherein the board bracket comprises a crossbar portion supporting a lowest surface of the puzzle board; and the at least one bracket storage member is disposed at a

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side of the crossbar portion away from the lowest surface of the puzzle board and is mounted to the crossbar portion.

2. The jigsaw puzzle table according to claim 1, further comprising a positioning fastener, connected to at least one of the board bracket and the lower support, wherein the positioning fastener is configured to secure the board bracket and the lower support to fix a height of the puzzle board supported by the board bracket and the lower support; the positioning fastener is further configured to release security between the board bracket and the lower support to enable the board bracket and the lower support to be at least partially movable relative to each other to adjust the supported height.

3. The jigsaw puzzle table according to claim 2, wherein the board bracket comprises support rods supporting the puzzle board, each of the support rods is sleeved in a respective support tube of the lower support; the jigsaw puzzle table further comprises a sleeve tube that sleeves an outside of the support rod that is exposed out of the support tube, and an outside of a portion of the support tube that sleeves the support rod; the positioning fastener is connected to the sleeve tube to fasten the support rod with the support tube.

4. The jigsaw puzzle table according to claim 3, wherein the positioning fastener comprises a fastening portion and an operating portion, the fastening portion is connected to the operating portion, the sleeve tube defines a fastening hole, the fastening portion is mounted in the fastening hole.

5. The jigsaw puzzle table according to claim 3, wherein the sleeve tube defines a first channel and a second channel communicated with the first channel, a shape of the first channel matches a shape of an outer surface of the support rod, and a shape of the second channel matches a shape of an outer surface of the support tube.

6. The jigsaw puzzle table according to claim 5, wherein a size of the second channel is not less than a size of the first channel, and the support rod passes through the second channel.

7. The jigsaw puzzle table according to claim 2, further comprising at least one board storage member; wherein the at least one board storage member is configured to store the puzzle pieces and is movably arranged inside the puzzle board; and the board bracket and the bracket rod are supported to a bottom side of the at least one board storage member.

8. The jigsaw puzzle table according to claim 7, wherein each of the at least one bracket storage member is movably arranged below the at least one board storage member; the bottom side of the at least one board storage member is higher than a top side of the at least one bracket storage member; the at least one bracket storage member and the at least one board storage member are located on different planes; each of the at least one board storage member is movable along a first direction, each of the at least one bracket storage member is movable along a second direction perpendicular to the first direction.

9. The jigsaw puzzle table according to claim 8, wherein a size of a storage cavity of each of the at least one board storage member is different from a size of a storage cavity of each of the at least one bracket storage member; and the at least one board storage member is arranged at a first side of the puzzle board, the at least one bracket storage member is arranged at a second side of the puzzle board different from the first side.

10. The jigsaw puzzle table according to claim 9, wherein a depth of the storage cavity of each of the at least one

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bracket storage member is larger than a depth of the storage cavity of each of the at least one board storage member.

11. The jigsaw puzzle table according to claim 9, wherein the at least one board storage member is arranged at a lateral side of the puzzle board, the at least one bracket storage member is arranged at a front side of the puzzle board.

12. The jigsaw puzzle table according to claim 11, wherein the puzzle board is arranged with a slide groove, the at least one board storage member is slidably connected to the slide groove; and the bracket rod is arranged with a support retaining edge, the at least one bracket storage member is slidably connected to the support retaining edge.

13. The jigsaw puzzle table according to claim 12, wherein the at least one bracket storage member has a pulling end configured to be pulled by a user to slide along the support retaining edge; the support retaining edge is arranged with a limiting block capable of supporting an end of each of the at least one bracket storage member away from the pulling end to prevent the at least one bracket storage member from moving excessively.

14. The jigsaw puzzle table according to claim 7, wherein the at least one bracket storage member is configured to be pulled out from a longer edge of the puzzle board; and the at least one board storage member is configured to be pulled out from a shorter edge of the puzzle board.

15. The jigsaw puzzle table according to claim 7, wherein, in a height direction of the jigsaw puzzle table, the bottom side of the at least one board storage member is higher than the top side of the at least one bracket storage member.

16. The jigsaw puzzle table according to claim 7, wherein the puzzle board comprises a puzzle carrying board, a support plate, and a movable support member; the puzzle carrying board is disposed above the support plate, the movable support member is disposed between the puzzle carrying board and the support plate; the puzzle carrying board is capable of switching, via the movable support member, between being placed flat on the support plate and being arranged inclined with respect to the support plate.

17. The jigsaw puzzle table according to claim 1, further comprising at least one board storage member; wherein the at least one board storage member is configured to store the puzzle pieces and is movably arranged inside the puzzle board; and the board bracket and the bracket rod are supported to a bottom side of the at least one board storage member.

18. A jigsaw puzzle table, comprising:
a puzzle board;

a board bracket, wherein the puzzle board is mounted on a mounting end of the board bracket; and

a lower support, wherein the board bracket is mounted between the puzzle board and the lower support, the board bracket and the lower support are configured to support the puzzle board cooperatively;

at least one bracket storage member and at least one board storage member, configured to store puzzle pieces, wherein the at least one bracket storage member is supported by and movably mounted on the board bracket; the at least one bracket storage member is movably arranged below the at least one board storage member; and the at least one bracket storage member and the at least one board storage member are pullable in different directions.

19. The jigsaw puzzle table according to claim 18, wherein a bottom side of the at least one board storage member is higher than a top side of the at least one bracket storage member; each of the at least one board storage member is movable along a first direction, each of the at

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least one bracket storage member is movable along a second direction perpendicular to the first direction.

20. The jigsaw puzzle table according to claim **1**, wherein the board bracket further comprises a plurality of bracket rods extending from the crossbar portion away from the puzzle board; the at least one bracket storage member is mounted to the plurality of bracket rods and the crossbar portion.

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